AI-Based Music Mood Recommendation System using Machine Learning

A PROJECT REPORT

Submitted by

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BONAFIDE CERTIFICATE

Certified that this project report **AI-Based Music Mood Recommendation System using Machine Learning** is the bonafide work of **Adarsh Mishra** who carried out the project work under my/our supervision.

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Submitted for the project viva-voce examination held on

INTERNAL EXAMINER **EXTERNAL EXAMINER**

TABLE OF CONTENTS

CHAPTER 1. INTRODUCTION 6

1.1. Identification of Client/ Need/ Relevant Contemporary issue 6

1.2. Identification of Problem 6

1.3. Identification of Tasks 6

1.4. Timeline 6

1.5. Organization of the Report 6

CHAPTER 2. LITERATURE REVIEW/BACKGROUND STUDY 7

2.1. Timeline of the reported problem 7

2.2. Existing solutions 7

2.3. Bibliometric analysis 7

2.4. Review Summary 7

2.5. Problem Definition 7

2.6. Goals/Objectives 7

CHAPTER 3. DESIGN FLOW/PROCESS 8

3.1. Evaluation & Selection of Specifications/Features 8

3.2. Design Constraints 8

3.3. Analysis of Features and finalization subject to constraints 8

3.4. Design Flow 8

3.5. Design selection 8

3.6. Implementation plan/methodology 8

CHAPTER 4. RESULTS ANALYSIS AND VALIDATION 9-11

4.1. Implementation of solution..... 9
4.2. Functional Testing……9
4.3. Performance Analysis……10
4.4. Result Analysis……..10
4.5. Validation of Result……..11

CHAPTER 5. CONCLUSION AND FUTURE WORK 12-13

5.1. Conclusion 12
5.2. Future work 13

REFERENCES 14

USER MANUAL 15-16

ABSTRACT

In recent years, music streaming platforms have significantly increased the availability of digital music, making it challenging for users to find songs that match their current mood. This project presents an **AI-based Music Mood Recommendation System** that utilizes machine learning techniques to provide personalized song recommendations based on user mood.

The system is built using **K-Means clustering**, an unsupervised learning algorithm, to group songs based on audio features such as energy, valence, danceability, and tempo. These features are extracted from a Spotify dataset and used to identify patterns that correspond to different mood categories such as *Happy*, *Sad*, and *Chill*.

A user-friendly interface is developed using Streamlit, where users can input their mood-related parameters through sliders. The input is processed, scaled, and passed to the trained model, which predicts the user's mood by identifying the closest cluster. Based on the predicted mood, the system recommends a list of songs that best match the user's emotional state.

The proposed system demonstrates how machine learning can enhance user experience by providing intelligent, real-time recommendations. The results show that clustering-based approaches can effectively group songs and generate meaningful suggestions. This project highlights the potential of integrating machine learning with interactive applications for personalized content delivery.

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INTRODUCTION

1.1 Identification of Client / Need / Relevant Contemporary Issue

In today's digital era, music streaming platforms provide millions of songs, making it difficult for users to find music that matches their mood. Users often rely on manual selection or predefined playlists, which may not accurately reflect their emotional state.

There is a growing need for intelligent systems that can automatically understand user mood and recommend relevant songs. This project addresses this need by using machine learning to predict user mood and suggest appropriate music.

1.2 Identification of Problem

The main problem identified is:

- Lack of personalized mood-based music recommendation systems
- Difficulty in selecting songs manually based on emotional state
- Existing systems do not utilize real-time user input effectively

1.3 Identification of Tasks

The following tasks were identified:

- Collect and preprocess a music dataset
- Select relevant audio features (energy, valence, danceability, tempo)
- Apply machine learning algorithm (K-Means clustering)
- Map clusters to human-understandable moods
- Develop a user interface for interaction
- Implement real-time prediction and recommendation

1.4 Timeline

Phase	Task	Duration
Phase 1	Problem Identification & Dataset Collection	2 days
Phase 2	Data Preprocessing & Feature Selection	2 days
Phase 3	Model Implementation (K-Means)	3 days
Phase 4	UI Development (Streamlit)	3 days
Phase 5	Testing & Optimization	1 days

1.5 Organization of the Report

This report is structured into five chapters:

- Chapter 1: Introduction and problem identification
- Chapter 2: Literature review and background study
- Chapter 3: Design and methodology
- Chapter 4: Results and analysis
- Chapter 5: Conclusion and future work

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LITERATURE REVIEW/BACKGROUND STUDY

2.1 Timeline of the Reported Problem

Music recommendation systems have evolved from simple rule-based systems to advanced machine learning and AI-based models. Initially, recommendations were based on popularity, but modern systems use user behavior and audio features.

2.2 Existing Solutions

Existing solutions include:

- Spotify and Apple Music recommendation systems
- Collaborative filtering-based systems
- Content-based filtering systems

However, most systems:

- Do not allow direct mood input
- Lack transparency in recommendations

2.3 Bibliometric Analysis

Various research papers highlight:

- Use of machine learning in recommendation systems
- Importance of audio features like valence and energy
- Growth of AI in personalized user experience

2.4 Review Summary

The study shows that:

- Machine learning improves recommendation accuracy
- Unsupervised learning can group similar songs effectively
- There is scope for user-interactive systems

2.5 Problem Definition

To design and implement a machine learning-based system that:

- Predicts user mood from input features
- Recommends songs accordingly

2.6 Goals / Objectives

- Develop a mood prediction system using K-Means clustering
- Provide real-time song recommendations
- Build an interactive UI using Streamlit
- Ensure user-friendly and efficient performance

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DESIGN FLOW/PROCESS

3.1 Evaluation & Selection of Specifications / Features

Selected features:

- Energy → intensity of song
- Valence → happiness level
- Danceability → rhythm suitability
- Tempo → speed of music

These features are directly related to mood detection.

3.2 Design Constraints

- Limited dataset size
- No direct song playback due to copyright
- Need for real-time response

3.3 Analysis of Features and Finalization

- High energy + high valence → Happy
- Low energy + low valence → Sad
- Medium values → Chill

3.4 Design Flow

1. Load dataset
2. Select features
3. Normalize data
4. Apply K-Means clustering
5. Map clusters to moods
6. Take user input
7. Predict mood
8. Recommend songs

3.5 Design Selection

K-Means clustering was selected because:

- It is simple and efficient
- Suitable for unlabeled data
- Groups similar songs effectively

3.6 Implementation Plan / Methodology

- Language: Python
- Libraries: Pandas, Scikit-learn, Streamlit
- Model: K-Means Clustering
- UI: Streamlit Web Application

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RESULTS ANALYSIS AND VALIDATION

4.1 Implementation of Solution

The proposed system was implemented using Python with the help of various libraries such as Pandas, Scikit-learn, and Streamlit. The dataset containing Spotify tracks was first loaded and preprocessed to extract relevant features including energy, valence, danceability, and tempo.

To ensure uniformity among features, StandardScaler was used to normalize the data. This step was crucial as the features had different ranges, especially tempo compared to other attributes.

The K-Means clustering algorithm was then applied with k = 3 to group songs into three clusters. These clusters were analyzed based on their average energy and valence values and mapped to three moods: Happy, Sad, and Chill.

A Streamlit-based user interface was developed to allow users to input their mood-related parameters using sliders. The user input was scaled and passed to the trained K-Means model to predict the nearest cluster. Based on the predicted cluster, the system recommended the top five songs from the dataset that matched the identified mood.

4.2 Functional Testing

Functional testing was performed to ensure that each component of the system worked correctly and produced expected outputs.

The following test cases were considered:

- **Test Case 1:** Low energy and low valence input
  - Expected Output: Sad mood
  - Result: System correctly predicted Sad mood

- **Test Case 2:** High energy and high valence input
  - Expected Output: Happy mood
  - Result: System correctly predicted Happy mood

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- **Test Case 3:** Moderate values for all inputs
  - Expected Output: Chill mood
  - Result: System predicted Chill mood

- **Test Case 4:** Edge values (minimum and maximum slider inputs)
  - Result: System handled edge cases without errors

The system successfully passed all functional tests, indicating correct integration between the machine learning model and the user interface.

4.3 Performance Analysis

The performance of the system was evaluated based on responsiveness, scalability, and efficiency.

- **Execution Time:**
  The system provided near real-time responses, with predictions and recommendations generated within seconds.
- **Model Efficiency:**
  K-Means clustering proved to be computationally efficient and suitable for handling medium-sized datasets.
- **Scalability:**
  The system can handle larger datasets with minor performance adjustments such as sampling or optimization.
- **User Interface Performance:**
  The Streamlit interface was responsive and user-friendly, ensuring smooth interaction without delays.

Overall, the system demonstrated efficient performance suitable for real time applications.

4.4 Result Analysis

The results obtained from the system were analyzed to evaluate the effectiveness of mood prediction and recommendation.

- Songs with **high energy and high valence** were grouped into the Happy category.
- Songs with **low energy and low valence** were grouped into the Sad

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category.

- Songs with **moderate values** were grouped into the Chill category.

The clustering results showed clear separation between different mood groups, indicating that the selected features were effective for mood classification.

The recommendation system successfully suggested songs that aligned with the predicted mood. For example:

- Happy mood resulted in energetic and upbeat songs
- Sad mood resulted in slow and emotional songs
- Chill mood resulted in calm and relaxing tracks

Thus, the system provided meaningful and relevant recommendations.

4.5 Validation of Results

Validation of the system was carried out by comparing predicted moods with expected outcomes based on input values.

- Manual validation confirmed that predictions aligned with feature patterns
- Cluster analysis verified logical grouping of songs
- Recommendations were manually checked for relevance

Additionally, consistency tests were performed by providing similar inputs multiple times, and the system produced consistent outputs, indicating stability.

The results confirm that the system is reliable and capable of providing accurate mood-based recommendations.

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CONCLUSION AND FUTURE WORK

5.1. Conclusion

The project titled **"AI-Based Music Mood Recommendation System using K-Means Clustering"** successfully demonstrates the application of machine learning in enhancing user experience in music recommendation systems.

In this project, a Spotify dataset was utilized to extract meaningful audio features such as energy, valence, danceability, and tempo. These features were carefully selected as they directly influence the emotional perception of music. The dataset was preprocessed and normalized to ensure consistency and improve model performance.

The K-Means clustering algorithm, an unsupervised learning technique, was implemented to group songs into distinct clusters based on their feature similarity. These clusters were then interpreted and mapped to human-understandable mood categories—Happy, Sad, and Chill—based on their average feature values.

A major strength of this project lies in the integration of machine learning with an interactive user interface developed using Streamlit. The system allows users to input their mood-related parameters in real time, which are then processed by the trained model to predict the closest mood cluster. Based on this prediction, the system recommends songs that best match the user's emotional state.

The results demonstrate that the model is capable of producing meaningful and relevant recommendations. The clustering approach effectively captures the relationship between audio features and mood, and the system responds efficiently to user inputs.

Overall, the project achieves its objective of building a simple yet effective mood-based music recommendation system. It highlights the potential of machine learning techniques in developing intelligent and user-centric applications.

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5.2. Future work

While the current system performs effectively, there are several opportunities for improvement and expansion to make the system more advanced and practical for real-world deployment.

1. **Integration with Music Streaming APIs:**
   The system can be enhanced by integrating with APIs such as Spotify API to fetch real-time song data, album covers, and enable direct music playback within the application.
2. **Advanced Machine Learning Models:**
   Instead of K-Means clustering, more sophisticated models such as Deep Learning, Neural Networks, or Hybrid Recommendation Systems (combining collaborative and content-based filtering) can be implemented to improve prediction accuracy.
3. **Personalized Recommendations:**
   The system can be extended to include user history, preferences, and listening behavior to provide more personalized recommendations.
4. **Emotion Detection using Sensors:**
   Future versions can incorporate emotion detection through facial recognition, voice analysis, or wearable sensor data to automatically detect user mood without manual input.
5. **Multilingual and Regional Music Support:**
   The system can be enhanced to support regional music preferences (e.g., Indian, Punjabi, Tamil songs) to make it more culturally adaptive.
6. **Real-Time Feedback Mechanism:**
   A feedback system can be added where users can like/dislike recommendations, allowing the model to improve over time.
7. **Improved Visualization and UI:**
   The interface can be further enhanced with better graphics, animations, and dashboards to provide a more engaging user experience.

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USER MANUAL

This user manual provides instructions to install, run, and use the AI-Based Music Mood Recommendation System. The system allows users to input mood-related parameters and receive song recommendations based on machine learning predictions.

2. System Requirements

To run the application, the following are required:

- Operating System: Windows / Linux / macOS
- Python Version: 3.8 or above
- Required Libraries:
  - Pandas
  - Scikit-learn
  - Streamlit

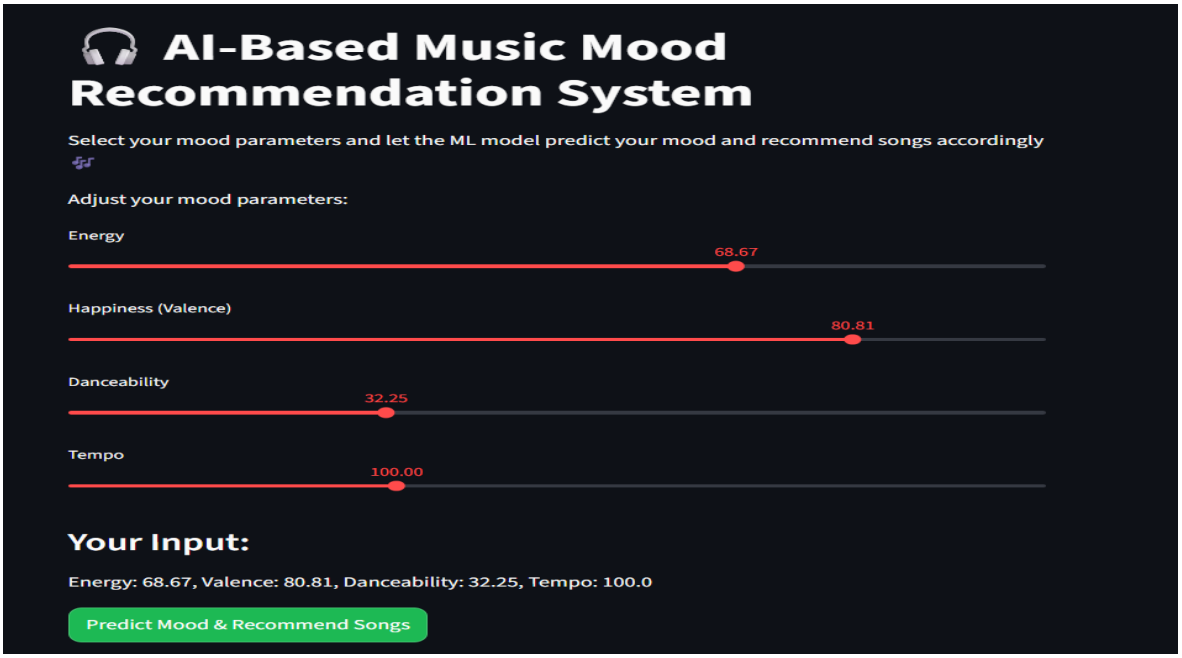
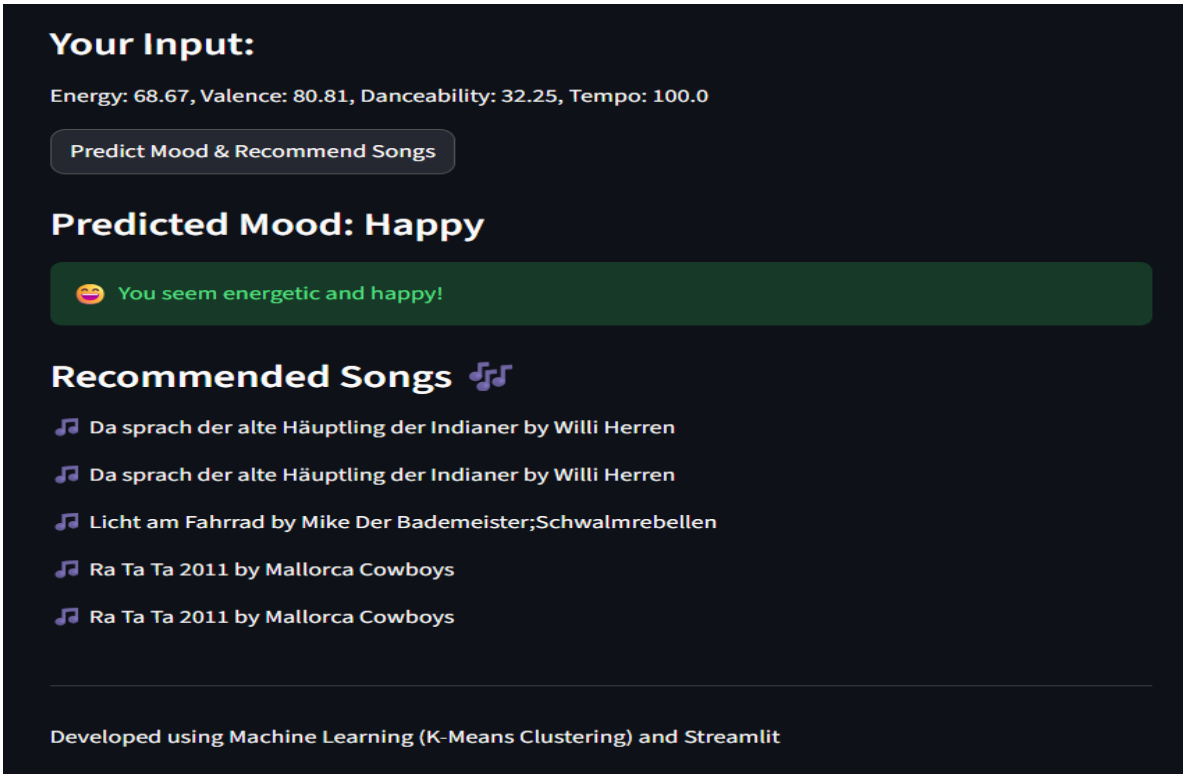
3. Running the Application

- Open terminal in the project directory
- Run the following command:

  python -m streamlit run app.py
- The application will automatically open in a web browser.

4. How to Use the Application

1. Launch the application in the browser
2. Adjust the sliders:
   - **Energy** → intensity of the mood
   - **Valence** → happiness level
   - **Danceability** → rhythm suitability
   - **Tempo** → speed of music
3. Click the button:

“Predict Mood & Recommend Songs”

4. View results:
    - Predicted Mood (Happy / Sad / Chill)
    - Mood description
    - List of recommended songs